

Name of Equipment	Cost in Lakhs	Quantity	Justification	Year	Department
Biochemistry Analyser (Integrated clinical chemistry and immunoassay analyzer) 1 No.	60	1	This instrument is essential for day-to-day research work involving biochemical analysis of serum samples from laboratory animals as a part of research in the areas of inflammation, cancer and metabolic disorders including diabetes, NASH, etc.	2026-27	Applied Biology
Real-Time PCR system (96 and 384 well module) with computer	20	1	Acquiring an RT-qPCR machine is crucial for precise, high-throughput gene expression analysis in in vitro and in vivo models. It supports studies on disease mechanisms, therapeutic responses, and biomarker discovery with high sensitivity and specificity. This versatile tool enhances the lab's capacity for advanced molecular biology research.	2026-27	Applied Biology
Microplate reader	20	1	A microplate reader is vital for high-throughput analysis of biochemical and cellular assays, offering versatile measurements of absorbance, fluorescence, and luminescence.	2026-27	Applied Biology

ARC Upgradation	60	1	Accelerating Rate Calorimetry (ARC) specializes in simulating worst-case adiabatic (no heat loss) conditions to study thermal runaway and hazards of self-accelerating exothermic reactions, unlike standard reaction calorimeters (like RC1 or DSC), which might operate under controlled cooling/isothermal conditions to measure heat flow for process optimization.	2026-27	CEPT
Melt Extrusion Unit	50	1	For the GFL project to extrude ion exchange membranes by melt extrusion at IICT	2026-27	CEPT
1-5 Kg Catalyst preparation units (Driers, Extruders, glass reactors, Low temperature cooling circulators, Furnaces)	100	1	Essential for the scale up of catalysts related industrial application Required for pilot scale studies of the industry sponsored projects	2026-27	C&FC
BCU-02 with 4 mm CPMAS/HRMAS probe and auto sampler	250	1	To cater the needs of internal as well as external samples, a solid-state NMR probe is required to integrate with existing 500 MHz NMR spectrometer. This probe is also suitable for analysing Hydrogels, Resins, Insoluble Polymers and Pharmaceutical Polymers. For better utilization of the instruments, an auto sampler is required.	2026-27	DASC
Differential Scanning Calorimeter (DSC) and Modulated DSC	90	1	We are in requirement of DSC along with MDSC.	2026-27	DASC
Scanning Electron Microscope	175	1	It is one of the essential equipment for material research	2026-27	DASC
Ash Fusion Tester	75	1	An Ash Fusion Tester is a specialized laboratory device used in coal analysis to determine the melting behaviour of coal ash at high temperatures. It is used to assess the temperature at which ash from coal starts to soften, fuse, or form a liquid, which is critical for understanding its behaviour during combustion in power plants and furnaces.	2026-27	DEEE

Dispersion Stability Analyzer	52	1	Formulation Equipment	2026-27	FAC
Particle Size Analyzer & Zeta Potential	45	1	Formulation Equipment	2026-27	FAC
Schrodinger Software (license for 5 years – 2026-2030)	25	1	Drug Design / Molecular modelling activities; Software license is categorized under capital head.	2026-27	NPMC
Circular Dichroism (CD)	70	1	For absolute stereochemistry of isolated natural products; to get valuable information of peptides, proteins and other chromophoric chiral molecules structural conformation. Proteinligand interaction, DrugDNA interaction and their subsequent change in the conformation, protein folding and aggregation studies can be studied in CD spectroscopy.	2026-27	NPMC
Pulsed NMR for Solid Fat Compositions	50	1	This equipment is used to measure the solid fat content of fat samples at different temperatures. The equipment we had in our department, earned considerable amount of revenues and we are getting many enquiries from the industries. The machine is not working now and its condition is beyond repairing.	2026-27	DOLST
Deep Freezer -80 0C freezer	15	1	Dedicated deep freezer for keeping tissue culture materials to be used in tissue culture lab (under development) in OLST to preserve tissue samples	2026-27	DOLST
Kjeldahl Equipment for Nitrogen Estimation	20	1	This is an equipment for Protein related work. As, we have some protein-based research projects and planned basic research on protein based	2026-27	DOLST
Gas generator	30	1	Generation of H ₂ , N ₂ and air for GC. Being an NABL accredited lab, gas generator is indispensable for uninterrupted operation of GCs.	2026-27	DOLST
FT-IR	70	2	To characterize compound FOR replacement of the existing one are older (more than 15 years)	2026-27	OSPC / PFM
GC-MS	75	1	To characterize and monitor a chemical reaction and mainly for mission mode projects. For	2026-27	OSPC

			replacement of the existing one are older (more than 15 years)		
Salt-spray unit	20	1	Salt-spray unit is important and routinely required for testing accelerated corrosion performance of coating and paint formulations received from both PFM and Industry. This unit at PFM is now more than 20 yrs old and spare parts are not available and therefore not in fully functional conditions	2026-27	PFM
QUV weathering	30	1	Used for measuring the accelerated weathering performance of the coating and paints it will be employed for testing samples from the IICT and industry. This unit at PFM is now more than 20 yrs old and spare parts are not available and therefore not in fully functional conditions	2026-27	PFM
Minor Equipment's	150		Stirrers, rotary evaporators, vacuum pumps, chillers, safety cabinets, Ice flake machines	2026-27	BDRM